Physical Model Breakdown

Construction Specifications

Second Skin Facade

Method: 3D Printing (additive)
Material: PLA
Colour: Burnt Orange (filament)

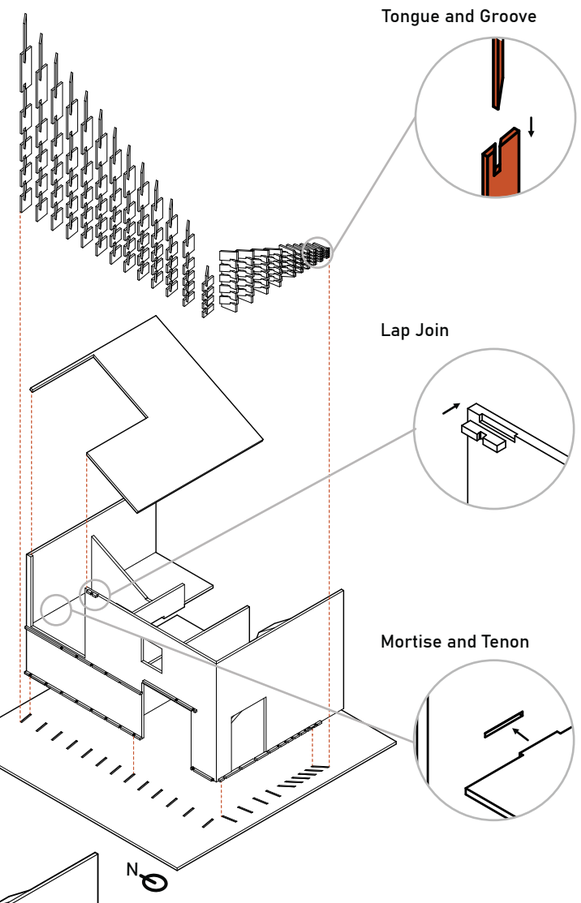
House Component

Method: Laser Cutting (subtractive)
Material: 3mm MDF
Colour: White (spray paint)

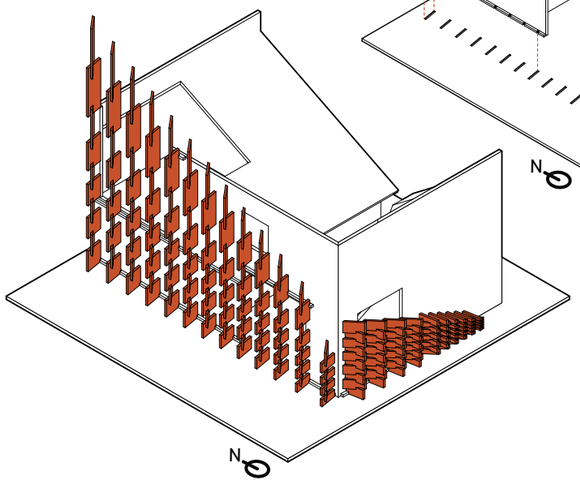
Base Component

Method: Laser Cutting (subtractive)
Material: 3mm MDF
Colour: Black (paint)

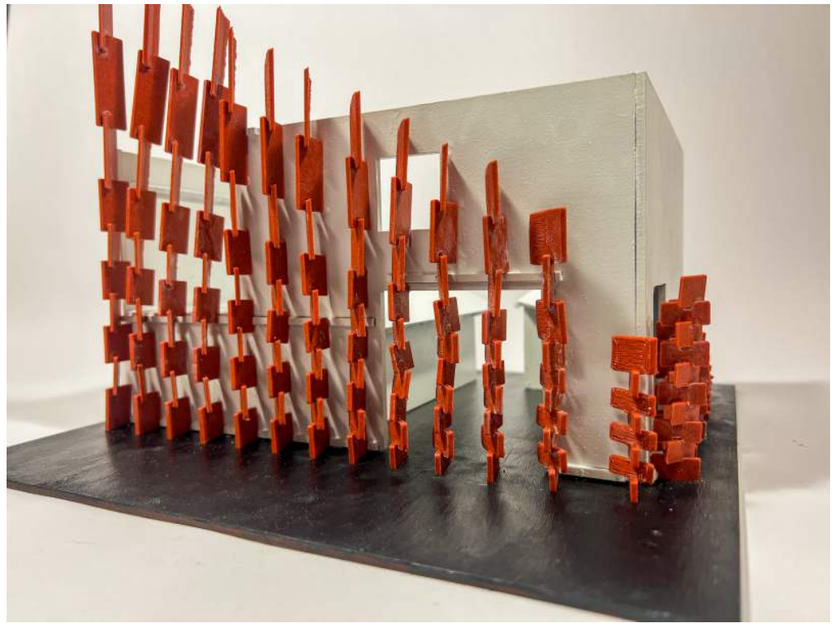
Detailed Views

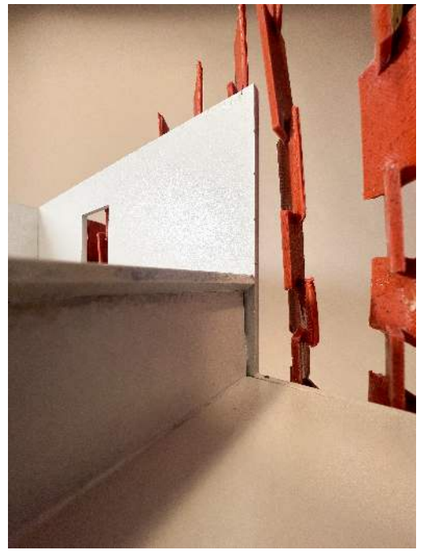


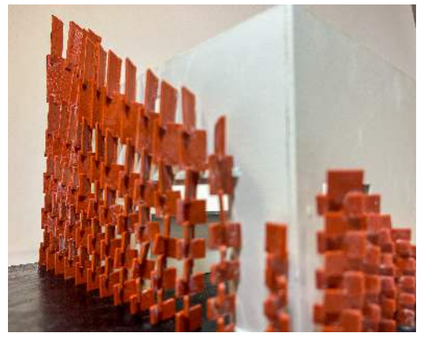
Final Model

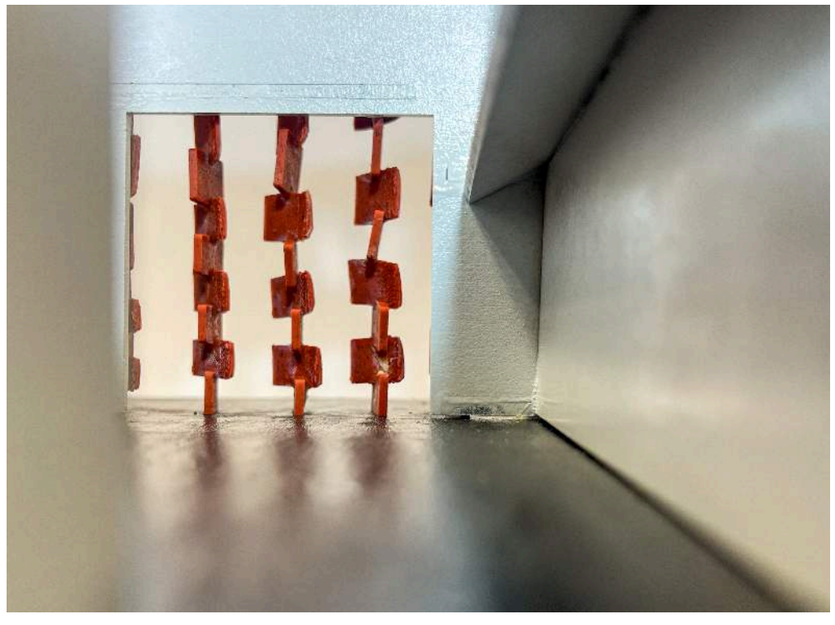


1:50 Model





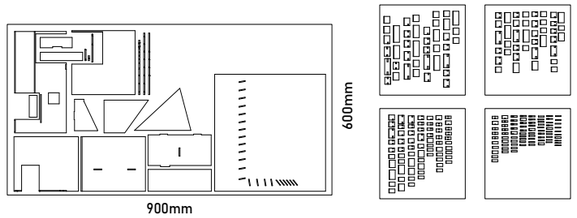






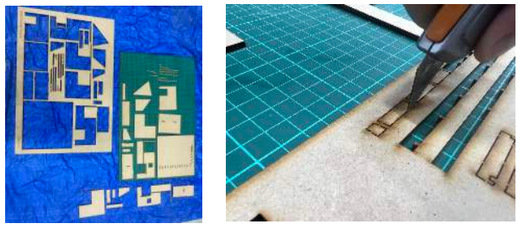
Construction Documentation

1. Extract and Nest Model Componentry



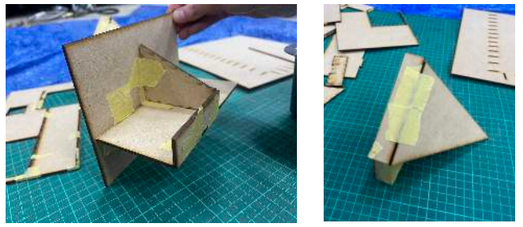
For both the 3D print and the laser cutting geometry, isolate the pieces and orient them to Z=0 in the Rhino file. Extract the line work for the laser cutting and nest the geometry on the 900mmx600mm cut area. Configure the 3D print geometry in their facade panel groups for ease of construction.

2. Receive and Cut Bridged Pieces from MDF



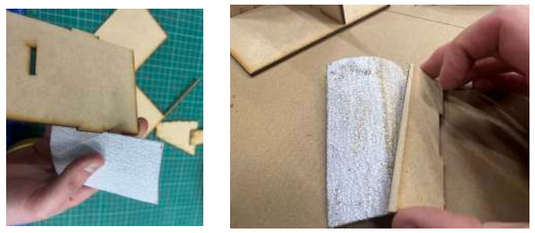
Remove the tape from the MDF laser-cut board and separate out the spare pieces cut from the extra space. Then taking a stanley knife, and following appropriate safety measures and techniques, remove the small bridged pieces. A few pieces broke during the process, so it was well planned to have spares.

3. Complete a Dry Assembly of House Geometry



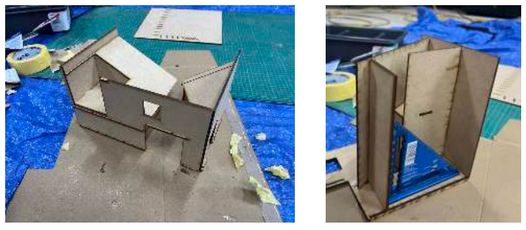
Using the house geometry pieces a dry assembly was first completed, minimising any irreversable mistakes that could occur while doing a wet assembly. This process also aided in discovering adjustments that were required to be made to the join geomerty.

4. Touch Up Edges for Joined Pieces



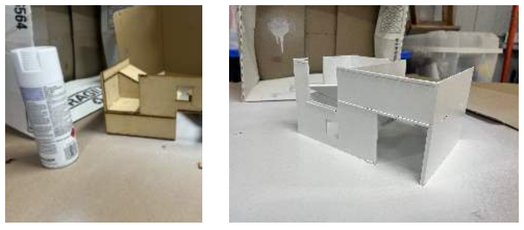
Small issues were discovered in the mortise and tenon join and the square edges of the roof supporting walls. For the mortise and tenon join, the componets were sanded down to achieve a better fit. For the walls, a mitred edge was sanded in to create a flush connection to the roof piece. 80 and 120 grit was used.

5. Glue Together House Geometry



Once the joinery errors were fixed, the next step was to glue the house geometry together. Superglue was used to speed up the process and a right angle triangle as used for more precise angles. The process began with the external geometry then working in, with the first glued pieces being the facade supports.

6. Spray Paint House Geometry



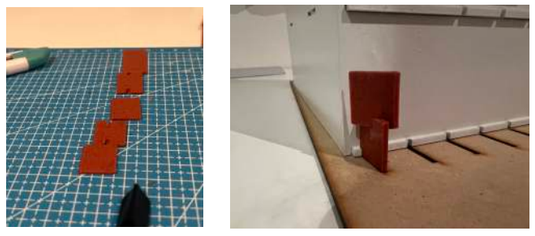
While spray painting the house all safety measures were followed, eg ventilation and PPE. Aesthetically I believe that the white contrasts well with the vibrate filament chosen for the facade. Functionally the white base allows for a more prominent visualisation of the shadow impact in the later photography.

7. Paint 300mmx300mm Base



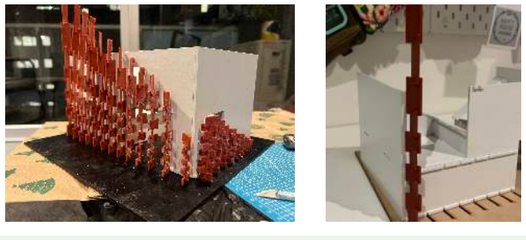
The base plate of the model was painted black with acrylic paint purely for aesthetic purposes, as I believe it contrasts the house and facade well. It also provides a more finished look than just plain MDF. A key consideration with painting the base was how the added paint thickness effects the joinery slots for the facade.

8. Group and Test Fit of 3D Printed Facade Pieces



Before gluing the facade together, organising and testing the individual pieces was crucial. With over 220 pieces, ensuring that the correct configuration was being followed and that the joins were clean was essential to produce a quality end result and to minimise confusion with the facade construction.

9. Glue Together Facade Geometry



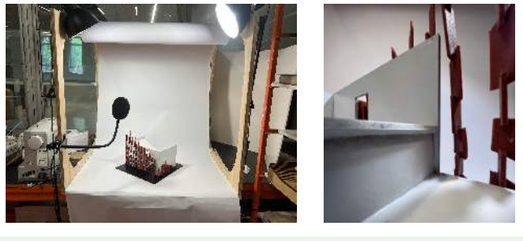
Each facade column was glued individually, moving from column A (left) to W (right). This was a very complex and time consuming process that required precision in the angled joining of the pieces. The only connection touch up required was some sanding to ensure closer fits for the pieces.

10. Clean Up Excess Glue on Facade Geometry



Following the glueing of the facade geometry with super glue some cleaning up was required. Using Acetone, any excess glue and glue stains that were left in the construction process were removed to produce and clean and quality end result.

11. Photograph on Photo Stage



Using the photo stage in Glyn Davis, quality images of the final model were produced. The process included; testing and captureing different lighting effects through the change of light source angles, light intensity and camera placement.

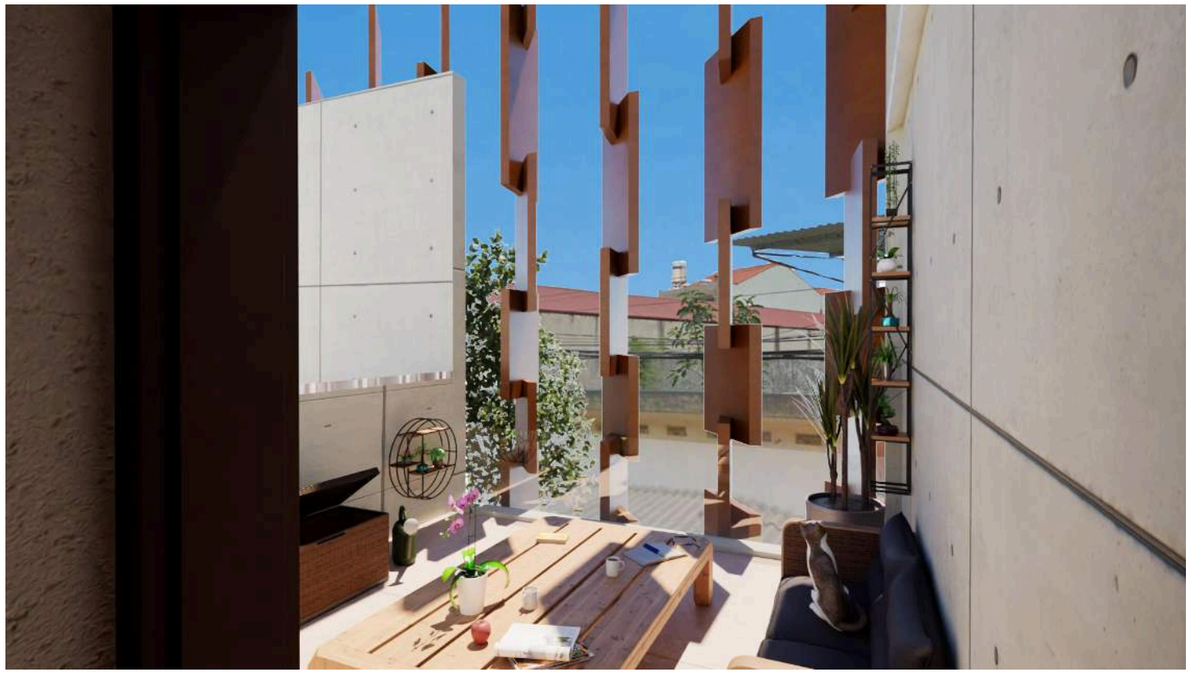
Partaking in the ever decreasing art of physical model making has provided me with a new perspective on the craft from a personal and commercial level. Personally, I find the process extremely beneficial in terms of both enjoyment and also in fully understanding the geometry in a tangible space. Having to construct the model forces an appreciation and consideration of the geometry in real world applications.

Reflecting on this through the commercial lens, I do understand why the frequency of this craft is declining from the perspective of time, cost and labour potentially not being outweighed by the benefits of creating a physcial model. However, as Tryba Architects, who still utilise physical model making, praises; physical models are " an expression of design philosophy, transformative ideas, and genuine artistry." (Tryba, 2022)

Renders



South and West Second Skin Facade, External, Afternoon



West Second Skin Facade, Internal, Afternoon

Rendering Process

1. Rhino to Twinmotion sync
2. Introduce and edit dynamic sky, for lighting
3. Model in additional context geometry
4. Map materials to geometry, and introduce doors
5. Populate 3D grass and add vegetation
6. Add figures and furniture
7. Improve context and colour correct in post-production



Post Production Detail

Colour correction filter layer utilised to improve the realism of the vegetation, matching it to the context image trees.



Background context layered in through the use of the Twinmotion skylayer exported image that was rasterised to a mask. Colour correction and refining brushes aided in creating the seemless context introduction.

Overlaying the image of the real site context on the additional mapped geometry of the concrete border. This was completed through layering different masked layers and changing the opacity to match the positioning.

Sky image layered positions using rasterised masks, colour referenced from context images taken in similar conditions and time to the render settings.



Use of masking layers and selection brushes to layer Google Earth site context. Perspective warp tool utilised to slightly adjust the image angles to align with the render.

Original Images for Photo-Montage

A. 

B. 

C. 

A. The original precident was used in the foreground and background of the external render to fully capture the design in context. Having the surrounding vegetation and buildings, along with the concrete pavement aids in the realism of the render. B&C. The Google Maps snapshots encaptulate what the terrace view would have been, furthering the render's presentation of interacting with surrounds.

External Render

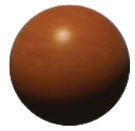
For the external render, capturing the limited context of the original precident's location was a priority. Intergating the render into the surrounds is crucial in order to present how the design interacts with the intended surroundings. This includes prominent vegitation, other buildings, roads and most prominent in this render the foot path and boundary concrete. Capturing the footpath context aids in presenting how the general public will interact with the exterior of the building. This is especially important with the DHY House due to its commerical operation as a coffee shop.

Some of Luxigon's more experimental concept design renders are exceptionally good examples of the benefits of rendering in context, due to their 'otherworldly' settings. By placing the design in clear context we can understand how we would interact with it (Megson, 2017). This was the aim of my external render, to represent how people interact with the house and how it interacts with the world.

Internal Render

The terrace was implemeted to further AHL Architect's design philosophy centred around communal spaces being created by promoting natural cooling (Abdel, 2023). This has been mirrored through the internal render's narrative. We see a space of gathering and community, alluded to through the selection of objects carefully placed in the space. By AHL Architect's design philophy, this shared space is generated by the natural cooling supplied by the shadows of the second skin.

Capturing this design impact so poignantly is a key benefit of renders. In industry it is clear that renders serve a clear visualisation communication purpose in presenting designs, but some firms adopt the philosophy that renders are more so a form of storytelling (Cutieru, 2021). I have intended to also adopt this philosophy for my internal render.



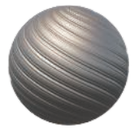
Red Cedar



Sandstone Tiles



Plaster Coating



Roof Covering Metal



Poured Concrete

References

Abdel, H. (2023, July 10). *DHY House / AHL architects*. October 22 2025. https://www.archdaily.com/1003649/dhy-house-ahl-architects

Cutieru, A. (2021, April 12). *The Different Uses of Renderings in Architecture.* October 25 2025. https://www.archdaily.com/959868/the-different-functions-of-a-rendering

Tryba Architects (2022, September 2). *Modeling the Future*. October 24, 2025. https://www.trybaarchitects.com/news/modeling-the-future

Megson, K. (2017). *Eric de Broches des Combes*; *Luxigon.* October 25, 2025. https://www.cladglobal.com/architecture-design-features?codeid=31601

Images (Render Context)

Google Maps (2019, July). *Street View.* https://www.google.com/maps/place/21%C2%B036'22.3%22N+105%C2%B050'23.8%22E/@21.606689,105.8403577,3a,90y,134.88h,90.61t/data=!3m7!1e1!3m5!1sEeN4HLfr4kF0addjMtHI6Q!2e0!6shttps:%2F%2Fstreetviewpixels-pa.googleapis.com%2Fv1%2Fthumbnail%3Fcb_client%3Dmaps_sv.tactile%26w%3D900%26h%3D600%26pitch%3D-0.6078459250250035%26panoid%3DEeN4HLfr4kF0addjMtHI6Q%26yaw%3D134.8822579918248!7i16384!8i8192!4m4!3m3!8m2!3d21.6061944!4d105.8399444?entry=ttu&g_ep=EgoyMDI1MTAyMC4wIKXMDSoASAFQAw%3D%3D

DHY House AHL Architects (2022). *Material - University of Melbourne.*https://canvas.lms.unimelb.edu.au/courses/214033/pages/assignments-and-submissions